

The maximum Laplacian ratio of a tree: a complete answer to a question of Brualdi and Goldwasser, kernel-checked in Lean

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26 September 2026 (preprint, not yet refereed)

Abstract

For a tree T with Laplacian $L(T)$, Brualdi and Goldwasser (1984) introduced the *Laplacian ratio* $\pi(T) = \text{per } L(T) / \prod_v \deg v$ and asked which tree on n vertices maximizes it. We answer the question for every n . For every $n \geq 4$ the maximum is attained by a *spider of cherry arms*: a centre carrying pendant paths of length two (cherries) and *arms*, an arm being a vertex that carries cherries. For $n \geq 492$ the centre carries only arms, almost all with five cherries, and the exceptional arms are fixed by $6(n-1) \bmod 11$ through an explicit rule; for $4 \leq n \leq 491$ the maximizer is given by an explicit table. Along the way we determine the exact exponential growth rate of the maximum, $\frac{64}{621}\rho^n \leq \max_{|T|=n} \pi(T) \leq 2\rho^{n-1}$ with $\rho = (621/64)^{1/11}$, and prove the sharp inequality behind it: every pendant subtree b has matching weight at most $\rho^{|b|}$, with equality exactly for the five-cherry arm. The whole argument is formalized in Lean 4 with Mathlib and checked by the Lean kernel, depending only on Lean’s three standard axioms. Large parts of it are certificates: exact data produced by untrusted generators and checked by the kernel. We describe the mathematics, the certificates, how the kernel checks them, and how to reproduce every step.

1 Introduction

Let T be a tree on $n \geq 2$ vertices, D its diagonal degree matrix, A its adjacency matrix and $L(T) = D - A$ its Laplacian. Brualdi and Goldwasser [1] studied the permanent of $L(T)$ and the *Laplacian ratio*

$$\pi(T) = \frac{\text{per } L(T)}{\prod_{v \in V(T)} \deg v}.$$

They showed $\pi(T) \geq 2$, with equality exactly for the star, and asked for the *maximum* over trees on n vertices. Wu, Dong and Lai [2] conjectured that the maximizer is a particular subdivided star. Pant [3] refuted that conjecture with explicit families of caterpillars whose spine vertices each carry many pendant paths of length two, and left the question open.

This paper answers it for every n .

Definition 1 (cherries, arms, spiders). A *cherry* is a pendant path of length two. For $j \geq 1$ an *arm* A_j is a vertex, adjacent to the centre, that carries j cherries and nothing else; it has $2j + 1$ vertices. We write A_0 for a leaf adjacent to the centre. A *spider of cherry arms* (in this paper, simply a *spider*) is a tree with a vertex, the *centre*, all of whose branches are leaves, cherries or arms. We write $C^c A_{j_1} A_{j_2} \cdots$ for the spider whose centre carries c cherries and arms $A_{j_1}, A_{j_2}, \dots$

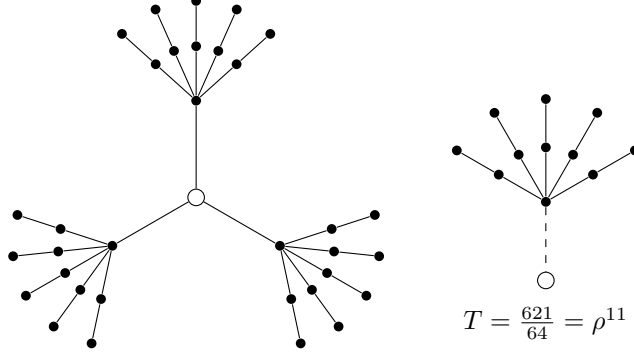


Figure 1: Left: the spider A_5^3 on $n = 34$ vertices (centre drawn open). Right: the five-cherry arm A_5 , an 11-vertex pendant subtree whose matching weight is exactly ρ^{11} ; it is the unique pendant subtree attaining the sharp bound of Theorem 5.

Theorem 2 (main theorem). *For every $n \geq 4$ the tree $\text{bgMax}(n)$ below has n vertices and maximizes π over all trees on n vertices.*

1. For $n \geq 492$, put $s = 6(n-1) \bmod 11$. The centre of $\text{bgMax}(n)$ carries only arms, and all of them are A_5 except:
 - $s \in \{1, 2, 3, 4\}$: s arms are A_6 ; except that for $s = 3$ with $n \leq 722$ and for $s = 4$ with $n \leq 2319$, instead $11-s$ arms are A_4 ;
 - $s \in \{5, \dots, 10\}$: $11-s$ arms are A_4 .
2. For $4 \leq n \leq 491$, $\text{bgMax}(n)$ is the spider listed in Appendix A.

For $n \leq 3$ there is only one tree on n vertices.

The exceptional counts are forced by vertex counting: $|A_4| = 9$, $|A_5| = 11$, $|A_6| = 13$, and $n-1 \equiv 2s \pmod{11}$, so replacing A_5 's by s copies of A_6 , or by $11-s$ copies of A_4 , uses up the residue. Figure 1 shows the building block.

Theorem 2 is the Lean theorem

```
theorem R3Cert.BGMaximizerAll.bg_maximizer_all (n : Nat) (h4 : 4 <= n) :
  usize (bgMax n) = n /\ forall t : UTree,
    usize t = n -> Aobj t <= Aobj (bgMax n)
```

together with its companion `bg_maximizer_all_perm`, which states the inequality literally for $\text{per}(L)/\prod \deg$ of the realized simple graphs (§6).

Why five cherries. The number $\rho = (621/64)^{1/11} \approx 1.22948$ is the exact exponential growth rate of the maximum.

Theorem 3 (growth rate). *For every $n \geq 1$, $\frac{64}{621} \rho^n \leq \max_{|T|=n} \pi(T) \leq 2 \rho^{n-1}$.*

The upper bound comes from a sharp inequality for pendant subtrees (Theorem 5): every pendant subtree b contributes a factor at most $\rho^{|b|}$, with equality only for the five-cherry arm. A maximizer therefore uses as many five-cherry arms as the arithmetic of n allows, and the whole difficulty is in the lower-order terms. Pant's families grow like $(3/2)^{n/2} \approx 1.2247^n$, just below ρ .

How the proof goes. The ratio is a weighted matching count, which a cavity recursion evaluates on rooted trees (§2). Rooting a tree at a vertex of maximum degree, we show that any tree which is not a spider is strictly beaten by some spider of the same size (§4). The spiders form an explicit family with a closed-form value, and we solve the optimization over it exactly (§5). Each step is split by size:

n	reduction to spiders	best spider
4–6	exact rational evaluation of every tree	
7–149	envelope certificate (degree ≤ 23) + high-degree theorem	table sweep
150–491	credited rate cells + high-degree theorem	table sweep
≥ 492	rate cells + high-degree theorem	exchange lemmas + 223 polynomial certificates

Status. Every step is formalized in Lean 4 with Mathlib [6] and checked by the Lean kernel; the headline theorems depend only on `propext`, `Classical.choice` and `Quot.sound`, with no `sorry`, no `native_decide` and no added axioms. The result has not yet been refereed. The kernel certifies the deductions; what a reader must still check is that the formal statement says what we claim it says, and §6 is written to make that check easy. The development was carried out with extensive help from an AI system (see the acknowledgements); for that reason we have tried to make every claim checkable by machine and every certificate reproducible from scratch.

2 The ratio as a matching sum, and the cavity recursion

Theorem 4 ([1, 3]). *For a tree T ,*

$$\text{per } L(T) = \sum_M \prod_{v \notin V(M)} \deg v, \quad \text{hence} \quad \pi(T) = \sum_M \prod_{uv \in M} \frac{1}{\deg u \deg v},$$

the sums running over all matchings M of T , including the empty one.

Proof. A term $\prod_i L_{i,\sigma(i)}$ of the permanent is nonzero only if every i is fixed by σ or sent to a neighbour. The non-trivial cycles of σ then run along edges of T ; since T has no cycles, they are transpositions along edges, that is, a matching M . A fixed point contributes $L_{ii} = \deg i$ and a matched edge contributes $L_{ij}L_{ji} = 1$. $\square$

In Lean this is `pi_eq_msum` (file `Matching.lean`), proved for any finite acyclic simple graph; the bridge `pi_utree` specializes it to the trees of the formalization.

The weighted matching sum is a monomer-dimer partition function in the sense of Heilmann and Lieb [4], with edge weights $1/(\deg u \deg v)$.

Planted branches. A *planted branch* b is a rooted tree whose root will be attached to one further vertex, its parent. All degrees are measured in the final tree, so the root of b has degree $d_b = (\text{number of children}) + 1$. Let T_b be the matching sum of Theorem 4 restricted to b , and T_b° the same sum over matchings that leave the root of b unmatched. Put $y_b = T_b^\circ / (d_b T_b)$. Splitting on whether, and to which child, the root is matched gives

$$T_b = \left(\prod_c T_c \right) \frac{d_b + R_b}{d_b}, \quad y_b = \frac{1}{d_b + R_b}, \quad R_b = \sum_c y_c, \quad (1)$$

the products and sums running over the children c of the root. A leaf has $T = 1$, $y = 1$. For a tree rooted at a vertex with k children,

$$\pi(T) = \left(\prod_c T_c \right) \left(1 + \frac{R}{k} \right), \quad R = \sum_c y_c. \quad (2)$$

Since (1) multiplies one factor per vertex, $\log T_b = \sum_{v \in b} \log(1 + R_v/d_v)$.

Atoms. Iterating (1): a cherry has $T = \frac{3}{2}$ and $y = \frac{1}{3}$. The arm A_j has $d = j + 1$ and $R = j/3$, so

$$T(A_j) = \left(\frac{3}{2} \right)^j \frac{4j + 3}{3(j + 1)}, \quad y(A_j) = \frac{3}{4j + 3}. \quad (3)$$

In particular $T(A_5) = (3/2)^5 \cdot \frac{23}{18} = \frac{621}{64}$. The per-vertex rates are

	A_1	A_2	A_3	A_4	A_5	A_6	A_7	cherry
$\frac{\log T}{\#\text{vertices}}$	0.18654	0.20232	0.20565	0.2064721	0.2065862	0.2064696	0.20628	0.2027326

and $\log T(A_j)/(2j + 1) \rightarrow \frac{1}{2} \log \frac{3}{2}$, the cherry value, as $j \rightarrow \infty$. The five-cherry arm is the unique best, with A_4 and A_6 just behind.

3 The sharp ceiling and the growth rate

Write $F^* = \frac{1}{11} \log \frac{621}{64} = \log \rho$ and, for a planted branch b with $|b|$ vertices, $\ell(b) = \log T_b - |b|F^*$.

Theorem 5 (sharp ceiling). *Every planted branch b satisfies $T_b \leq \rho^{|b|}$, that is $\ell(b) \leq 0$, with equality if and only if b is the five-cherry arm A_5 .*

In Lean: `R3Cert.BGSCL.bg.sharp`.

Outline. Telescoping. By the remark after (2), $\ell(b) = \sum_{v \in b} e_v$ with $e_v = \log(1 + R_v/d_v) - F^*$. Suppose $\varrho \geq 0$ is a function of a vertex's local state with

$$e_v + \varrho(v) \leq \sum_{c \text{ child of } v} \varrho(c) \quad \text{for every vertex } v. \quad (4)$$

Summing (4) over b telescopes to $\ell(b) \leq -\varrho(\text{root}) \leq 0$. In the language of ergodic optimization ϱ is a *subaction*.

The witness. We use an explicit subaction ϱ_{wit} depending on a vertex's number of children and on its message y : it equals F^* at a leaf, is affine in y at vertices with one, two or three children, and vanishes at vertices with four or more. Three constraints are tight, and they are exactly the three vertex types of A_5 : leaf, cherry vertex, and the arm vertex with five cherries. (That the extremal structure is a finite periodic pattern is what makes a finite piecewise-affine subaction possible; compare [5].) Their consistency is the identity $(3/2)^5 \cdot \frac{23}{18} = \frac{621}{64}$, that is $27 \cdot 23 = 621$. Every other instance of (4) holds with rational slack. The verification splits into finitely many cells by degree (among them 35 cells for vertices with four children, and a tail wrapper for high degree), each an inequality between rational envelopes and a rational enclosure of a logarithm (Taylor or atanh series with explicit remainder); see `BGSCLSubaction*.lean`.

Equality. If $\ell(b) = 0$ then every cell along b is tight. The root cell is tight only at degree six, and at degree six only when all five children are cherries: one non-cherry child makes the cell strict, using the strict inequality $2F^* - \log \frac{3}{2} < \frac{1}{96}$. That inequality is where strictness enters; at degree three the non-strict version is exactly tight. $\square$

Proof of Theorem 3. Root T anywhere. In (2) each child c is a planted branch, so $\prod_c T_c \leq \rho^{n-1}$ by Theorem 5; and each $y_c \leq 1$, so $1 + R/k \leq 2$. For the lower bound, take a centre carrying $K = \lfloor (n-1)/11 \rfloor$ arms A_5 , and fill the remaining at most ten vertices with cherries and at most one leaf; each $T \geq 1$, so $\pi \geq \rho^{11K} \geq \frac{64}{621} \rho^n$. In Lean: `R3Cert.Step3.bg_max_growth`. $\square$

4 Every maximizer is a spider

Root a tree at a vertex of maximum degree k ; call its children $b_1, \dots, b_k$. Every vertex then has at most $k-1$ children. Rerooting preserves π and the vertex count (`RerootRel`, `exists_maxRooted`). The *atoms* are the leaf, the cherry and the arms A_j ; the tree is a spider centred at the root exactly when every b_i is an atom. Taking logarithms in (2),

$$\log \pi - (n-1)F^* = \sum_{i=1}^k \ell(b_i) + \log \left(1 + \frac{S}{k}\right), \quad S = \sum_i y_{b_i}. \quad (5)$$

By Theorem 5 every term of the sum is ≤ 0 . The task is to show that a non-atom child costs more than any possible gain in the last term, measured against the best spider of the same size. We compare with explicit spiders: the A_5/A_4 spider (with $q \equiv 5(n-1) \pmod{11}$ arms A_4 and the rest A_5) satisfies

$$\log \pi - (n-1)F^* \geq \log \frac{26}{23} - \frac{10}{960}, \quad (6)$$

because A_4 has $y = \frac{3}{19} > \frac{3}{23} = y(A_5)$ and each A_4 costs little.

High degree ($k \geq 24$). Use the tangent of $\log(1 + S/k)$ at $S/k = 3/23$, which prices each unit of message at $\mu = \frac{23}{26k} \leq \frac{23}{624}$. For $V_\mu(b) = \ell(b) + \mu y_b$ we show that on this range of μ the arm A_5 is the envelope of V_μ over all branches, and every non-atom lies below it by at least $\frac{1}{75}$ (the observed minimum is 0.01452, at a vertex carrying five cherries and an A_4). This reduces, through the subaction, to three per-vertex statements, each proved with rational tangent certificates and degree-six Taylor enclosures of $\log$. Hence a max-degree rooting with a non-atom child has $\log \pi - (n-1)F^* \leq \log \frac{26}{23} - \frac{1}{75}$, below (6). This settles $k \geq 24$ for $n \geq 91$ (`spider_dominates_highDegree_uncond`); for $25 \leq n \leq 90$ we compare with the table spider instead, by a kernel-checked inequality at each n (`BGHighDegreeSmall.highDegree_small`).

Low degree, large n ($k \leq 23$, $n \geq 492$). Here we trade the subaction's slack for a *linear deficit*. A *rate cell* with degree cap D and rate α is a per-vertex inequality which, telescoped as in (4), yields

$$\ell(b) + \varrho_{\text{wit}}(b) \leq -\alpha|b| \quad \text{for every non-atom branch } b \text{ with at most } D-1 \text{ children per vertex.}$$

We use $D = 23$ with $\alpha = \frac{1}{3700}$ (root degree 6 to 23), and $D = 5, 4, 3, 2$ with $\alpha = \frac{1}{2100}, \frac{1}{660}, \frac{1}{420}, \frac{1}{135}$ (root degree 5, 4, 3, 2). There are 32 rate cells and 22 root cells; the tightest margin is $2.8 \cdot 10^{-4}$. Then $\log \pi - (n-1)F^* \leq R_k - \alpha(n-1)$ with $R_k \in [0.208, 0.319]$, which falls below (6) once $n-1 \geq 491$ (`spider_dominates_lowDegree_uncond`).

Low degree, $150 \leq n \leq 491$. The same scheme with two refinements: in a tree of maximum degree $\leq D$ every message satisfies $y \geq 1/(2D-1)$, which narrows the per-cell ranges; and non-atom branches carry a *credit* κ depending on their number of children, $\ell(b) + \varrho_{\text{wit}}(b) \leq -\alpha|b| - \kappa(b)$, which they pass to the root. The root is then bounded by an explicit knapsack over atoms, certified separately for each pair (k, n) (`BGSpiderMidCells*`, `BGSpiderMidRoot*`; `spider_dominates_of_maxDegreeRoot.150`).

Low degree, $7 \leq n \leq 149$: the envelope certificate. For small n the linear deficit is too weak, and we instead compute a Bellman envelope. Fix a rational rate f close to F^* (it cancels between a tree and a spider of the same size) and a grid of prices μ_g . With $V_\mu(b) = \log T_b - |b|f + \mu y_b$, the tangent of $x \mapsto \log x + \mu/x$ bounds V_μ at a vertex by a constant plus $\sum_c V_\nu(c)$ at a new price ν (`BGEnvCert.Tangent`), and a max-plus knapsack over child sizes bounds the sum. A table $w(m, g)$, indexed by size m and grid point g , is a valid envelope if it satisfies finitely many such inequalities; the envelope theorem (`BGEnvCert.Envelope`) then gives $V_{\mu_g}(b) \leq w(|b|, g)$ for every branch of size m with at most C children per vertex, and a sharper table for non-atoms. The root inequality with at least one non-atom child is checked against certified lower bounds for explicit spiders (`BGEnvCert.Spider`).

All tables are stored in fixed point (34 fractional bits) and packed into lanes of large natural numbers, and the checks are Boolean functions that the kernel evaluates. We cover root degrees $2 \leq k \leq 23$ with caps $C \in \{1, \dots, 8, 12, 16, 22\}$; roots of degree ≥ 24 are handled by the high-degree theorems. The certificate is sharded into 57 Lean files (§7) and assembled by `BGEnvCert.AssembleK` into `G149.envcertK`; the result is `maximizer_is_spider_small`.

$4 \leq n \leq 6$. Every tree is enumerated by a fuel-bounded structural recursion and its value computed exactly over $\mathbb{Q}$ (`BGMaximizerTiny`).

Theorem 6. *For every $n \geq 7$, every tree on n vertices that maximizes π is, after rerooting at a vertex of maximum degree, a spider. For $n \geq 492$ every tree on n vertices is weakly beaten by a spider on n vertices.*

In Lean: `maximizer_is_spider_small` ($7 \leq n \leq 149$), `BGMax.maximizer_is_spider_150` ($n \geq 150$), `BGMax.bg_spider_reduction` ($n \geq 492$).

5 The best spider

A spider is described up to isomorphism by the multiset of its centre's children. By (2) and (3) its value has the closed form

$$F = \left(\frac{3}{2}\right)^{c+\sum j} \prod_{\text{arms}} \alpha_j \cdot \left(1 + \frac{c/3 + \sum b_j}{D}\right), \quad \alpha_j = \frac{4j+3}{3(j+1)}, \quad b_j = \frac{3}{4j+3},$$

where c is the number of cherries, the product and sum run over the arms (leaves being A_0 , with $\alpha_0 = b_0 = 1$), and D is the degree of the centre. In Lean, F is a function on lists of children over $\mathbb{Q}$, and `Aobj_spiderU` identifies it with π of the realized tree.

Lemma 7 (balance). *Put $Q_c(x, y) = \alpha_x \alpha_y (c + b_x + b_y)$. For integers $k, t \geq 0$,*

$$Q_c(k+t+1, k+1) - Q_c(k+t+2, k) = \frac{(t+1)((c-3)(8k+4t+15)+3)}{9(k+1)(k+2)(k+t+2)(k+t+3)}.$$

Consequently, if $D \geq 3$, moving a cherry from an arm to an arm at least two smaller strictly increases F , and in every maximizer with $D \geq 3$ all arm sizes (leaves included) differ by at most one.

Proof. The move keeps n , D and the power of $\frac{3}{2}$, and multiplies F by $Q_c(\text{new})/Q_c(\text{old})$ with $c = D + R_0 \geq 3$, where R_0 is the message sum of the other children. The identity is a polynomial identity (`balance_identity`). $\square$

Two further exact exchanges are used: two leaves are never better than one cherry (`F_leafpair_lt`), and a general local-exchange criterion that is affine in R_0 and so reduces to two quadratics in D_0 (`F_exchange_lt`).

$4 \leq n \leq 491$. By Lemma 7, a maximizer with $D \geq 3$ is a permutation of a canonical configuration (c, s, a, b) : c cherries, a arms A_s and b arms A_{s+1} . Its value has an exact integer numerator and denominator. For each n the kernel compares every canonical configuration on n vertices, and every configuration with $D \leq 2$, against the table entry, by cross-multiplied inequalities of natural numbers (`BGSpiderTable.spider_opt_table`). The sweep is split into 89 chunk files of about five sizes each. At $n = 21$ two non-isomorphic spiders tie (C^{10} and $C^3 A_3^2$); the table records $C^3 A_3^2$.

$n \geq 492$: **structure.** Let l be a maximizer of F with $n \geq 492$. Then (Lean: `structProp_492`) the centre has at least three children; there is no arm A_j with $j \geq 13$ (split it into A_{j-11} and two A_5); none with $7 \leq j \leq 12$ (exchange eleven A_j for $2j + 1$ copies of A_5 , and a global comparison with the rule spider); at most eight cherries (nine cherries lose to two A_4); no arm with $j \leq 3$; so every arm is A_4 , A_5 or A_6 , A_4 and A_6 do not occur together, and there are at most ten of each. The global comparison uses $(\prod T)^2 = (3/2)^{n-1} \prod h$ with $h(\text{cherry}) = 1$ and $h(A_j) = \frac{2}{3}\alpha_j^2$; its tightest instance is $24.45 < 28.43$.

$n \geq 492$: **the candidates.** A candidate is determined by its counts (c, k_4, k_5, k_6) , and $F = \Phi(c, k_4, k_5, k_6)$ with $\Phi = (\frac{3}{2})^c G_4^{k_4} G_5^{k_5} G_6^{k_6} (1 + \frac{c/3+k_4 B_4+k_5 B_5+k_6 B_6}{c+k_4+k_5+k_6})$, $G_4 = \frac{513}{80}$, $G_5 = \frac{621}{64}$, $G_6 = \frac{6561}{448}$, $B_4 = \frac{3}{19}$, $B_5 = \frac{3}{23}$, $B_6 = \frac{1}{9}$. The triple (c, k_4, k_6) fixes the residue class s and hence the rule winner's counts; after dividing out a common power of G_5 , the comparison $\Phi(\text{candidate}) \leq \Phi(\text{winner})$ becomes $P(m) \geq 0$ for an explicit quadratic P in the common number m of five-cherry arms. There are 189 triples and 223 certificates: 221 write $P(L + t)$ with nonnegative coefficients in t , and two (the competitions that decide the thresholds 722 and 2319) write it in the Bernstein basis on a bounded interval. Lean checks each identity with `ring` and each sign with `norm_num` (`BGSpiderCand.candProp_492`).

Together these prove Theorem 2: Theorem 6 reduces to spiders, and this section finds the best spider.

6 The formal statement

This section is for a reader who wants to check that the Lean theorem means what Theorem 2 says. The relevant definitions are short.

- `UTree` is the type of finite rooted trees, built by `node` from a list of subtrees, and `usize` counts vertices. Every finite tree arises, rooted anywhere.
- `Step3.realize (dtRealize t)` turns t into an explicit edge list, and `aGraph` into a Mathlib `SimpleGraph` on its vertex type.
- `lapl G` is the matrix with $\deg i$ on the diagonal, -1 on edges and 0 elsewhere, and `Matrix.permanent` is Mathlib's permanent.
- `Aobj t` is defined as the matching sum of the realized tree; `pi_utree` proves

$$(\text{lapl } (\text{aGraph } \dots)).\text{permanent} / (\text{prod over } v \text{ of degree } v) = \text{Aobj } t.$$

- `bgMax n` is `spiderU (tab n)` for $n \leq 491$ and `spiderU (W n)` otherwise, where `tab` reads the table `BGSpiderTableData.tabData` and `W` implements the rule of Theorem 2.

So `bg_maximizer_all_perm` states, for every $n \geq 4$ and every tree t on n vertices, that $\text{per } L / \prod \deg \text{ of } t$ is at most that of `bgMax(n)`, with the permanent and the degrees read off the actual graph. The file `AxiomGuard.lean` prints the axioms of the 17 headline theorems.

What is not claimed in Lean. The Lean theorem gives *a* maximizer. That it is unique up to isomorphism (except at $n = 21$) is established by exact computation for $n \leq 491$ (below) and for $n \geq 2320$ by an exact rational certificate, but uniqueness is not formalized.

7 Certificates and Telperion

Most of the proof’s bulk is data: envelope tables, rate-cell constants, table rows, polynomial coefficients. We produce it with generators written in Python using exact integer and rational arithmetic, and we never trust the generators: each certificate is a Lean file whose checks the kernel evaluates, so a wrong certificate is simply a failed build.

The certificates are packaged with Telperion, a pipeline built on exactly that division of labour: an untrusted *emitter* produces a certificate, a small adapter renders it as Lean, and the Lean kernel is the only trusted component. For each certificate family the repository contains the generator, a *frozen* copy of its output with hashes, and a driver that regenerates the family from scratch and compares byte for byte, both with the frozen copy and with the files in the formalization:

family	Lean files	content
<code>spider_table</code>	91	table rows and the per- n sweep, $n \leq 491$
<code>envcert_g149</code>	57	envelope tables, tangents, spider bounds, $7 \leq n \leq 149$
<code>mid_cells</code>	45	credited rate cells and root knapsacks, $150 \leq n \leq 491$
<code>cand_certs</code>	10	the 223 polynomial certificates, $n \geq 492$

Making the kernel do it. Three facts about the Lean kernel shaped the certificates. (i) It does not reduce definitions by well-founded recursion efficiently, so every recursive check is structural, with explicit fuel. (ii) It does not share the evaluation of repeated argument subterms, so a loop state reused twice is evaluated twice; we emit literal constants and force evaluation of natural numbers before passing them on (`BGEnvCert.Force`). (iii) Arithmetic in $\mathbb{Q}$ is slow, so large checks use natural numbers and fixed point, and only small ones use $\mathbb{Q}$. We use `decide +kernel` and never `native_decide`, so the compiler is not trusted. Because the kernel keeps every intermediate term of a declaration in memory, large checks are split into many small theorems and many files; the heaviest files still need 15–35 GiB each.

8 Independent computations

None of the following is needed for the Lean proof; they are independent checks.

- *Certified search*, $n \leq 491$. A bottom-up search over all rooted trees, with no degree cap, storing every quantity as an outward-rounded interval and discarding a partial state only when it is provably strictly beaten in every completion. Surviving candidates are re-evaluated

in exact rational arithmetic. For every $n \leq 491$ the exact maximum equals the table value, and *every* maximizing tree is a spider. (952s on one core; `certificates/checks`.)

- *Brute force*. Exact enumeration of all non-isomorphic trees agrees for $n \leq 18$.
- *Spider optimization*. An exhaustive exact optimization over the spider family for $n \leq 2400$, together with an exact rational certificate for $n \geq 2320$, agrees with the table and the rule; an exact Pareto dynamic programme over the family with no balance assumption agrees for $n \leq 600$.

9 Reproducing the results

The repository <https://github.com/DrMurphyIsIn/brualdi-goldwasser> contains the Lean project (Lean v4.32.0, Mathlib), the certificate generators with their frozen outputs, and the Telperion engine.

```
cd formalization && lake exe cache get && lake build -j 4
lake env lean AxiomGuard.lean
certificates/verify.sh          # regenerate all certificates, compare
certificates/verify.sh --full  # also the certified search, n <= 491
```

The full build takes several CPU-hours and needs a machine with at least 64 GiB of memory.

10 Limitations

- The result has not been refereed. The Lean kernel checks the deductions; the correspondence between the formal definitions and the mathematical statement (§6) is what a human reviewer should check.
- The proof is long and mostly computational. We do not have a short conceptual proof that the maximizer is a spider, only per-vertex inequalities verified by certificate, and the thresholds ($n = 150, 491, 492$) are artefacts of the method.
- Uniqueness is not formalized.
- The trusted base is the Lean kernel, Mathlib’s definitions of the permanent and of simple graphs, and our definitions listed in §6.

Acknowledgements

The formalization, the certificate generators and much of the exploration were developed with extensive assistance from Claude, an AI system made by Anthropic, working under the author’s direction. The author takes responsibility for the content. The correctness of the formal proof rests on the Lean kernel, not on either.

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A The maximizers for $4 \leq n \leq 491$

Notation: C^c is c cherries at the centre, A_j^n is m arms with j cherries, L a leaf at the centre. The third column is π/ρ^{n-1} , which by Theorem 3 lies in $[\frac{64}{621}\rho, 2]$. At $n = 21$, C^{10} ties with the listed tree. Generated from `BGSpiderTableData.lean` by `paper/gen_table.py`.

n	maximizer	π/ρ^{n-1}	n	maximizer	π/ρ^{n-1}	n	maximizer	π/ρ^{n-1}
4	A_1	1.3452	167	$C^3 A_5^8 A_4^8$	1.1378	330	$A_5^{25} A_4^6$	1.1288
5	C^2	1.3129	168	$C^3 A_5^{13} A_4^2$	1.1383	331	A_5^{30}	1.1304
6	$C A_1$	1.2904	169	$C^3 A_5^9 A_4^7$	1.1376	332	$A_5^{26} A_4^5$	1.1291
7	C^3	1.3029	170	$C^4 A_5^{13} A_4^2$	1.1380	333	$C A_5^{30}$	1.1283
8	$C^2 A_1$	1.2657	171	$C^3 A_5^{10} A_4^6$	1.1373	334	$A_5^{27} A_4^4$	1.1293
9	C^4	1.2929	172	$C^4 A_5^{14} A_4$	1.1378	335	$A_5^{23} A_4^9$	1.1277
10	$C^2 A_2$	1.2658	173	$C^3 A_5^{11} A_4^5$	1.1371	336	$A_5^{28} A_4^3$	1.1296
11	C^5	1.2829	174	$C^4 A_5^{15}$	1.1375	337	$A_5^{24} A_4^8$	1.1280
12	$C^2 A_3$	1.2609	175	$C^3 A_5^{12} A_4^4$	1.1369	338	$A_5^{29} A_4^2$	1.1299
13	C^6	1.2731	176	$C^5 A_5^{15}$	1.1365	339	$A_5^{25} A_4^7$	1.1283
14	$C^3 A_3$	1.2620	177	$C^3 A_5^{13} A_4^3$	1.1366	340	$A_5^{30} A_4$	1.1302
15	C^7	1.2633	178	$C^2 A_5^{10} A_4^7$	1.1359	341	$A_5^{26} A_4^6$	1.1286
16	$C^3 A_4$	1.2587	179	$C^3 A_5^{14} A_4^2$	1.1364	342	A_5^{31}	1.1304
17	C^8	1.2536	180	$C^2 A_5^{11} A_4^6$	1.1357	343	$A_5^{27} A_4^5$	1.1289
18	$C^4 A_4$	1.2575	181	$C^3 A_5^{15} A_4$	1.1361	344	$A_6 A_5^{30}$	1.1281
19	C^9	1.2440	182	$C^2 A_5^{12} A_4^5$	1.1354	345	$A_5^{28} A_4^4$	1.1292
20	$C^4 A_5$	1.2535	183	$C^3 A_5^{16}$	1.1359	346	$A_5^{24} A_4^9$	1.1275
21	$C^3 A_3^2$	1.2344	184	$C^3 A_5^{12} A_4^5$	1.1352	347	$A_5^{29} A_4^3$	1.1295
22	$C^5 A_5$	1.2504	185	$C^4 A_5^{16}$	1.1355	348	$A_5^{25} A_4^8$	1.1278
23	$C^4 A_3^2$	1.2335	186	$C^3 A_5^{13} A_4^4$	1.1350	349	$A_5^{30} A_4^2$	1.1298
24	$C^5 A_6$	1.2454	187	$C^2 A_5^{10} A_4^8$	1.1346	350	$A_5^{26} A_4^7$	1.1281
25	$C^4 A_4 A_3$	1.2336	188	$C^3 A_5^{14} A_4^3$	1.1349	351	$A_5^{31} A_4$	1.1301
26	$C^6 A_6$	1.2409	189	$C^2 A_5^{11} A_4^7$	1.1344	352	$A_5^{27} A_4^6$	1.1285
27	$C^4 A_4^2$	1.2336	190	$C^3 A_5^{15} A_4^2$	1.1347	353	A_5^{32}	1.1304
28	$C^6 A_7$	1.2351	191	$C^2 A_5^{12} A_4^6$	1.1342	354	$A_5^{28} A_4^5$	1.1288
29	$C^5 A_4^2$	1.2322	192	$C^3 A_5^{16} A_4$	1.1345	355	$A_6 A_5^{31}$	1.1281
30	$C^7 A_7$	1.2296	193	$C^2 A_5^{13} A_4^5$	1.1340	356	$A_5^{29} A_4^4$	1.1291
31	$C^5 A_5 A_4$	1.2297	194	$C^3 A_5^{17}$	1.1343	357	$A_5^{25} A_4^9$	1.1272

n	maximizer	π/ρ^{n-1}	n	maximizer	π/ρ^{n-1}	n	maximizer	π/ρ^{n-1}
32	$C^7 A_8$	1.2232	195	$C^2 A_5^{14} A_4^4$	1.1339	358	$A_5^{30} A_4^3$	1.1294
33	$C^5 A_5^2$	1.2271	196	$C^4 A_5^{17}$	1.1336	359	$A_5^{26} A_4^8$	1.1276
34	$C^8 A_8$	1.2171	197	$C^2 A_5^{15} A_4^3$	1.1337	360	$A_5^{31} A_4^2$	1.1298
35	$C^6 A_5^2$	1.2246	198	$C A_5^{12} A_4^7$	1.1332	361	$A_5^{27} A_4^7$	1.1280
36	$C^4 A_4^3$	1.2162	199	$C^2 A_5^{16} A_4^2$	1.1335	362	$A_5^{32} A_4$	1.1301
37	$C^7 A_5^2$	1.2206	200	$C^2 A_5^{12} A_4^7$	1.1330	363	$A_5^{28} A_4^6$	1.1283
38	$C^5 A_4^3$	1.2159	201	$C^2 A_5^{17} A_4$	1.1333	364	A_5^{33}	1.1304
39	$C^7 A_6 A_5$	1.2167	202	$C^2 A_5^{13} A_4^6$	1.1329	365	$A_5^{29} A_4^5$	1.1287
40	$C^5 A_5 A_4^2$	1.2138	203	$C^2 A_5^{18}$	1.1331	366	$A_6 A_5^{32}$	1.1281
41	$C^7 A_6^2$	1.2128	204	$C^2 A_5^{14} A_4^5$	1.1328	367	$A_5^{30} A_4^4$	1.1290
42	$C^6 A_5 A_4^2$	1.2118	205	$C^3 A_5^{18}$	1.1329	368	$A_5^{26} A_4^9$	1.1270
43	$C^8 A_6^2$	1.2082	206	$C^2 A_5^{15} A_4^4$	1.1327	369	$A_5^{31} A_4^3$	1.1294
44	$C^6 A_5^2 A_4$	1.2102	207	$C A_5^{12} A_4^8$	1.1324	370	$A_5^{27} A_4^8$	1.1274
45	$C^8 A_7 A_6$	1.2031	208	$C^2 A_5^{16} A_4^3$	1.1325	371	$A_5^{32} A_4^2$	1.1297
46	$C^6 A_5^3$	1.2085	209	$C A_5^{13} A_4^7$	1.1323	372	$A_5^{28} A_4^7$	1.1278
47	$C^5 A_4^4$	1.2030	210	$C^2 A_5^{17} A_4^2$	1.1324	373	$A_5^{33} A_4$	1.1301
48	$C^7 A_5^3$	1.2056	211	$C A_5^{14} A_4^6$	1.1321	374	$A_5^{29} A_4^6$	1.1282
49	$C^5 A_5 A_4^3$	1.2013	212	$C^2 A_5^{18} A_4$	1.1323	375	A_5^{34}	1.1304
50	$C^7 A_6 A_5^2$	1.2020	213	$C A_5^{15} A_4^5$	1.1320	376	$A_5^{30} A_4^5$	1.1286
51	$C^6 A_5 A_4^3$	1.1998	214	$C^2 A_5^{19}$	1.1322	377	$A_6 A_5^{33}$	1.1282
52	$C^7 A_6^2 A_5$	1.1983	215	$C A_5^{16} A_4^4$	1.1319	378	$A_5^{31} A_4^4$	1.1289
53	$C^6 A_5^2 A_4^2$	1.1984	216	$C^3 A_5^{19}$	1.1316	379	$A_5^{27} A_4^9$	1.1268
54	$C^7 A_6^3$	1.1947	217	$C A_5^{17} A_4^3$	1.1317	380	$A_5^{32} A_4^3$	1.1293
55	$C^6 A_5^3 A_4$	1.1970	218	$C A_5^{13} A_4^8$	1.1315	381	$A_5^{28} A_4^8$	1.1272
56	$C^5 A_4^5$	1.1924	219	$C A_5^{18} A_4^2$	1.1316	382	$A_5^{33} A_4^2$	1.1297
57	$C^6 A_5^4$	1.1956	220	$C A_5^{14} A_4^7$	1.1314	383	$A_5^{29} A_4^7$	1.1276
58	$C^5 A_5 A_4^4$	1.1910	221	$C A_5^{19} A_4$	1.1315	384	$A_5^{34} A_4$	1.1301
59	$C^7 A_5^4$	1.1934	222	$C A_5^{15} A_4^6$	1.1314	385	$A_5^{30} A_4^6$	1.1280
60	$C^6 A_5 A_4^4$	1.1897	223	$C^2 A_5^{19} A_4$	1.1314	386	A_5^{35}	1.1304
61	$C^8 A_5^4$	1.1900	224	$C A_5^{16} A_4^5$	1.1313	387	$A_5^{31} A_4^5$	1.1284
62	$C^6 A_5^2 A_4^3$	1.1885	225	$C^2 A_5^{20}$	1.1313	388	$A_6 A_5^{34}$	1.1282
63	$C^8 A_6 A_5^3$	1.1867	226	$C A_5^{17} A_4^4$	1.1312	389	$A_5^{32} A_4^4$	1.1288
64	$C^6 A_5^3 A_4^2$	1.1873	227	$A_5^{14} A_4^8$	1.1311	390	$A_5^{28} A_4^9$	1.1267
65	$C^5 A_4^6$	1.1835	228	$C A_5^{18} A_4^3$	1.1311	391	$A_5^{33} A_4^3$	1.1292
66	$C^6 A_5^4 A_4$	1.1862	229	$A_5^{15} A_4^7$	1.1310	392	$A_5^{29} A_4^8$	1.1271
67	$C^5 A_5 A_4^5$	1.1824	230	$C A_5^{19} A_4^2$	1.1311	393	$A_5^{34} A_4^2$	1.1296
68	$C^6 A_5^5$	1.1850	231	$A_5^{16} A_4^6$	1.1309	394	$A_5^{30} A_4^7$	1.1275
69	$C^5 A_5^2 A_4^4$	1.1812	232	$C A_5^{20} A_4$	1.1310	395	$A_5^{35} A_4$	1.1300
70	$C^7 A_5^5$	1.1832	233	$A_5^{17} A_4^5$	1.1309	396	$A_5^{31} A_4^6$	1.1279
71	$C^6 A_5^2 A_4^4$	1.1801	234	$C A_5^{21}$	1.1309	397	A_5^{36}	1.1304
72	$C^8 A_5^5$	1.1802	235	$A_5^{18} A_4^4$	1.1308	398	$A_5^{32} A_4^5$	1.1283
73	$C^6 A_5^3 A_4^3$	1.1791	236	$A_5^{14} A_4^9$	1.1307	399	$A_6 A_5^{35}$	1.1282
74	$C^8 A_6 A_5^4$	1.1771	237	$A_5^{19} A_4^3$	1.1307	400	$A_5^{33} A_4^4$	1.1288
75	$C^6 A_5^4 A_4^2$	1.1782	238	$A_5^{15} A_4^8$	1.1307	401	$A_5^{29} A_4^9$	1.1265
76	$C^5 A_5 A_4^6$	1.1750	239	$A_5^{20} A_4^2$	1.1306	402	$A_5^{34} A_4^3$	1.1292
77	$C^6 A_5^5 A_4$	1.1772	240	$A_5^{16} A_4^7$	1.1306	403	$A_5^{30} A_4^8$	1.1269
78	$C^5 A_5^2 A_4^5$	1.1740	241	$C A_5^{20} A_4^2$	1.1306	404	$A_5^{35} A_4^2$	1.1296
79	$C^6 A_5^6$	1.1762	242	$A_5^{17} A_4^6$	1.1306	405	$A_5^{31} A_4^7$	1.1274
80	$C^5 A_5^3 A_4^4$	1.1730	243	$C A_5^{21} A_4$	1.1305	406	$A_5^{36} A_4$	1.1300
81	$C^7 A_5^6$	1.1746	244	$A_5^{18} A_4^5$	1.1306	407	$A_5^{32} A_4^6$	1.1278
82	$C^5 A_5^4 A_4^3$	1.1720	245	$C A_5^{22}$	1.1305	408	A_5^{37}	1.1304
83	$C^8 A_5^6$	1.1718	246	$A_5^{19} A_4^4$	1.1306	409	$A_5^{33} A_4^5$	1.1282

n	maximizer	π/ρ^{n-1}	n	maximizer	π/ρ^{n-1}	n	maximizer	π/ρ^{n-1}
84	$C^6 A_5^4 A_4^3$	1.1712	247	$A_5^{15} A_4^9$	1.1302	410	$A_6 A_5^{36}$	1.1282
85	$C^8 A_6 A_5^2 A_4^2$	1.1688	248	$A_5^{20} A_4^3$	1.1305	411	$A_5^{34} A_4^4$	1.1287
86	$C^6 A_5^5 A_4^2$	1.1704	249	$A_5^{16} A_4^8$	1.1303	412	$A_5^{30} A_4^9$	1.1263
87	$C^5 A_5^5 A_4^6$	1.1678	250	$A_5^{21} A_4^2$	1.1305	413	$A_5^{35} A_4^3$	1.1291
88	$C^6 A_5^3 A_4^5$	1.1696	251	$A_5^{17} A_4^7$	1.1303	414	$A_5^{31} A_4^8$	1.1268
89	$C^5 A_5^3 A_4^5$	1.1669	252	$A_5^{22} A_4^2$	1.1305	415	$A_5^{36} A_4^2$	1.1296
90	$C^6 A_5^7 A_4^4$	1.1688	253	$A_5^{18} A_4^6$	1.1303	416	$A_5^{32} A_4^7$	1.1272
91	$C^5 A_5^4 A_4^4$	1.1661	254	A_5^{23}	1.1304	417	$A_5^{37} A_4^4$	1.1300
92	$C^7 A_5^7 A_4^3$	1.1672	255	$A_5^{19} A_4^5$	1.1303	418	$A_5^{33} A_4^6$	1.1277
93	$C^5 A_5^2 A_4^3$	1.1653	256	$C A_5^{23}$	1.1301	419	A_5^{38}	1.1304
94	$C^8 A_5^6 A_4^2$	1.1646	257	$A_5^{20} A_4^4$	1.1304	420	$A_5^{34} A_4^5$	1.1282
95	$C^5 A_5^3 A_4^7$	1.1645	258	$A_5^{16} A_4^9$	1.1298	421	$A_6 A_5^{37}$	1.1282
96	$C^5 A_5^3 A_4^7$	1.1623	259	$A_5^{21} A_4^3$	1.1304	422	$A_5^{35} A_4^4$	1.1286
97	$C^6 A_5^6 A_4^2$	1.1637	260	$A_5^{17} A_4^8$	1.1299	423	$A_5^{31} A_4^9$	1.1262
98	$C^5 A_5^3 A_4^6$	1.1616	261	$A_5^{22} A_4^2$	1.1304	424	$A_5^{36} A_4^3$	1.1291
99	$C^6 A_5^7 A_4^4$	1.1631	262	$A_5^{18} A_4^7$	1.1300	425	$A_5^{32} A_4^8$	1.1266
100	$C^5 A_5^4 A_4^5$	1.1609	263	$A_5^{23} A_4^2$	1.1304	426	$A_5^{37} A_4^2$	1.1295
101	$C^6 A_5^8 A_4^3$	1.1624	264	$A_5^{19} A_4^6$	1.1300	427	$A_5^{33} A_4^7$	1.1271
102	$C^5 A_5^8 A_4^3$	1.1602	265	A_5^{24}	1.1304	428	$A_5^{38} A_4^6$	1.1300
103	$C^7 A_5^8 A_4^3$	1.1608	266	$A_5^{20} A_4^5$	1.1301	429	$A_5^{34} A_4^6$	1.1276
104	$C^5 A_5^6 A_4^3$	1.1595	267	$C A_5^{24}$	1.1298	430	A_5^{39}	1.1304
105	$C^8 A_5^8 A_4^2$	1.1582	268	$A_5^{21} A_4^4$	1.1302	431	$A_5^{35} A_4^5$	1.1281
106	$C^5 A_5^7 A_4^2$	1.1588	269	$A_5^{17} A_4^9$	1.1295	432	$A_6 A_5^{38}$	1.1282
107	$C^5 A_5^3 A_4^7$	1.1568	270	$A_5^{22} A_4^3$	1.1302	433	$A_5^{36} A_4^4$	1.1285
108	$C^5 A_5^8 A_4^2$	1.1581	271	$A_5^{18} A_4^8$	1.1296	434	$A_6 A_5^{37}$	1.1260
109	$C^5 A_5^4 A_4^6$	1.1562	272	$A_5^{23} A_4^2$	1.1303	435	$A_5^{37} A_4^3$	1.1290
110	$C^5 A_5^9 A_4^5$	1.1574	273	$A_5^{19} A_4^7$	1.1297	436	$A_5^{33} A_4^8$	1.1265
111	$C^5 A_5^5 A_4^5$	1.1556	274	$A_5^{24} A_4^2$	1.1304	437	$A_5^{38} A_4^2$	1.1295
112	$C^6 A_5^9 A_4^4$	1.1568	275	$A_5^{20} A_4^6$	1.1298	438	$A_5^{34} A_4^7$	1.1270
113	$C^5 A_5^6 A_4^4$	1.1551	276	A_5^{25}	1.1304	439	$A_5^{39} A_4^6$	1.1300
114	$C^7 A_5^9 A_4^3$	1.1552	277	$A_5^{21} A_4^5$	1.1299	440	$A_5^{35} A_4^6$	1.1275
115	$C^5 A_5^7 A_4^3$	1.1545	278	$C A_5^{25}$	1.1295	441	A_5^{40}	1.1304
116	$C^4 A_5^4 A_4^7$	1.1527	279	$A_5^{22} A_4^4$	1.1300	442	$A_5^{36} A_4^5$	1.1280
117	$C^5 A_5^8 A_4^2$	1.1539	280	$A_5^{18} A_4^9$	1.1291	443	$A_6 A_5^{39}$	1.1282
118	$C^4 A_5^5 A_4^6$	1.1521	281	$A_5^{23} A_4^3$	1.1301	444	$A_5^{37} A_4^4$	1.1285
119	$C^5 A_5^9 A_4^2$	1.1534	282	$A_5^{19} A_4^8$	1.1293	445	$A_6 A_5^{38}$	1.1261
120	$C^4 A_5^6 A_4^5$	1.1515	283	$A_5^{24} A_4^2$	1.1302	446	$A_5^{38} A_4^3$	1.1290
121	$C^5 A_5^{10} A_4^4$	1.1528	284	$A_5^{20} A_4^7$	1.1294	447	$A_5^{34} A_4^8$	1.1264
122	$C^5 A_5^6 A_4^5$	1.1510	285	$A_5^{25} A_4^2$	1.1303	448	$A_5^{39} A_4^2$	1.1295
123	$C^6 A_5^{10} A_4^4$	1.1520	286	$A_5^{21} A_4^6$	1.1296	449	$A_5^{35} A_4^7$	1.1269
124	$C^5 A_5^7 A_4^4$	1.1506	287	A_5^{26}	1.1304	450	$A_5^{40} A_4^6$	1.1299
125	$C^7 A_5^{10} A_4^3$	1.1502	288	$A_5^{22} A_4^5$	1.1297	451	$A_5^{36} A_4^6$	1.1274
126	$C^5 A_5^8 A_4^3$	1.1501	289	$C A_5^{26}$	1.1292	452	A_5^{41}	1.1304
127	$C^4 A_5^5 A_4^7$	1.1487	290	$A_5^{23} A_4^4$	1.1299	453	$A_5^{37} A_4^5$	1.1279
128	$C^5 A_5^9 A_4^2$	1.1496	291	$A_5^{19} A_4^9$	1.1288	454	$A_6 A_5^{40}$	1.1283
129	$C^4 A_5^6 A_4^6$	1.1482	292	$A_5^{24} A_4^3$	1.1300	455	$A_5^{38} A_4^4$	1.1284
130	$C^5 A_5^{10} A_4^4$	1.1492	293	$A_5^{20} A_4^8$	1.1290	456	$A_6 A_5^{39}$	1.1261
131	$C^4 A_5^7 A_4^4$	1.1477	294	$A_5^{25} A_4^2$	1.1301	457	$A_5^{39} A_4^3$	1.1289
132	$C^5 A_5^{11} A_4^3$	1.1487	295	$A_5^{21} A_4^7$	1.1292	458	$A_5^{35} A_4^8$	1.1263
133	$C^4 A_5^8 A_4^4$	1.1472	296	$A_5^{26} A_4^2$	1.1303	459	$A_5^{40} A_4^2$	1.1294
134	$C^6 A_5^{11} A_4^3$	1.1477	297	$A_5^{22} A_4^6$	1.1293	460	$A_5^{36} A_4^7$	1.1268
135	$C^4 A_5^9 A_4^3$	1.1468	298	A_5^{27}	1.1304	461	$A_5^{41} A_4^6$	1.1299

n	maximizer	π/ρ^{n-1}	n	maximizer	π/ρ^{n-1}	n	maximizer	π/ρ^{n-1}
136	$C^7 A_5^{11}$	1.1458	299	$A_5^{23} A_4^5$	1.1295	462	$A_5^{37} A_4^6$	1.1273
137	$C^4 A_5^{10} A_4^2$	1.1463	300	$C A_5^{27}$	1.1289	463	A_5^{42}	1.1304
138	$C^4 A_5^6 A_4^7$	1.1451	301	$A_5^{24} A_4^4$	1.1297	464	$A_5^{38} A_4^5$	1.1278
139	$C^5 A_5^{10} A_4^2$	1.1459	302	$A_5^{20} A_4^9$	1.1285	465	$A_6 A_5^{41}$	1.1283
140	$C^4 A_5^7 A_4^6$	1.1447	303	$A_5^{25} A_4^3$	1.1299	466	$A_5^{39} A_4^4$	1.1283
141	$C^5 A_5^{11} A_4$	1.1455	304	$A_5^{21} A_4^8$	1.1287	467	$A_6^2 A_5^{40}$	1.1261
142	$C^4 A_5^8 A_4^5$	1.1444	305	$A_5^{26} A_4^2$	1.1301	468	$A_5^{40} A_4^3$	1.1289
143	$C^5 A_5^{12}$	1.1451	306	$A_5^{22} A_4^7$	1.1289	469	$A_5^{36} A_4^8$	1.1261
144	$C^4 A_5^9 A_4^4$	1.1440	307	$A_5^{27} A_4$	1.1303	470	$A_5^{41} A_4^2$	1.1294
145	$C^6 A_5^{12}$	1.1439	308	$A_5^{23} A_4^6$	1.1291	471	$A_5^{37} A_4^7$	1.1267
146	$C^4 A_5^{10} A_4^3$	1.1436	309	A_5^{28}	1.1304	472	$A_5^{42} A_4$	1.1299
147	$C^3 A_5^7 A_4^7$	1.1424	310	$A_5^{24} A_4^5$	1.1294	473	$A_5^{38} A_4^6$	1.1272
148	$C^4 A_5^{11} A_4^2$	1.1432	311	$C A_5^{28}$	1.1287	474	A_5^{43}	1.1304
149	$C^3 A_5^8 A_4^6$	1.1420	312	$A_5^{25} A_4^4$	1.1296	475	$A_5^{39} A_4^5$	1.1278
150	$C^4 A_5^{12} A_4$	1.1428	313	$A_5^{21} A_4^9$	1.1282	476	$A_6 A_5^{42}$	1.1283
151	$C^4 A_5^8 A_4^6$	1.1417	314	$A_5^{26} A_4^3$	1.1298	477	$A_5^{40} A_4^4$	1.1283
152	$C^4 A_5^{13}$	1.1424	315	$A_5^{22} A_4^8$	1.1285	478	$A_6^2 A_5^{41}$	1.1261
153	$C^4 A_5^9 A_4^5$	1.1414	316	$A_5^{27} A_4^2$	1.1300	479	$A_5^{41} A_4^3$	1.1288
154	$C^5 A_5^{13}$	1.1419	317	$A_5^{23} A_4^7$	1.1287	480	$A_5^{37} A_4^8$	1.1260
155	$C^4 A_5^{10} A_4^4$	1.1411	318	$A_5^{28} A_4$	1.1302	481	$A_5^{42} A_4^2$	1.1294
156	$C^6 A_5^{13}$	1.1405	319	$A_5^{24} A_4^6$	1.1290	482	$A_5^{38} A_4^7$	1.1266
157	$C^4 A_5^{11} A_4^3$	1.1408	320	A_5^{29}	1.1304	483	$A_5^{43} A_4$	1.1299
158	$C^3 A_5^8 A_4^7$	1.1399	321	$A_5^{25} A_4^5$	1.1292	484	$A_5^{39} A_4^6$	1.1271
159	$C^4 A_5^{12} A_4^2$	1.1405	322	$C A_5^{29}$	1.1285	485	A_5^{44}	1.1304
160	$C^3 A_5^9 A_4^6$	1.1395	323	$A_5^{26} A_4^4$	1.1294	486	$A_5^{40} A_4^5$	1.1277
161	$C^4 A_5^{13} A_4$	1.1401	324	$A_5^{22} A_4^9$	1.1279	487	$A_6 A_5^{43}$	1.1283
162	$C^3 A_5^{10} A_4^5$	1.1392	325	$A_5^{27} A_4^3$	1.1297	488	$A_5^{41} A_4^4$	1.1282
163	$C^4 A_5^{14}$	1.1398	326	$A_5^{23} A_4^8$	1.1282	489	$A_6^2 A_5^{42}$	1.1261
164	$C^3 A_5^{11} A_4^4$	1.1389	327	$A_5^{28} A_4^2$	1.1299	490	$A_5^{42} A_4^3$	1.1288
165	$C^5 A_5^{14}$	1.1391	328	$A_5^{24} A_4^7$	1.1285	491	$A_5^{38} A_4^8$	1.1259
166	$C^3 A_5^{12} A_4^3$	1.1386	329	$A_5^{29} A_4$	1.1302			

$$n \text{ maximizer } \pi/\rho^{n-1} \quad n \text{ maximizer } \pi/\rho^{n-1} \quad n \text{ maximizer } \pi/\rho^{n-1}$$
